



Definite Integration: RACE-01

Single Correct Questions :

1. Suppose that $F(x)$ is an antiderivative of $f(x) = \frac{\sin x}{x}, x > 0$ then $\int_1^3 \frac{\sin 2x}{x} dx$ can be expressed as-
- (A) $F(6) - F(2)$ (B) $\frac{1}{2}(F(6) - F(2))$ (C) $\frac{1}{2}(F(3) - F(1))$ (D) $2(F(6) - F(2))$

Integer Type Questions :

Evaluate :

2. Let $[t]$ denote the greatest integer $\leq t$. Then the value of $8 \cdot \int_{-\frac{1}{2}}^1 ([2x] + |x|) dx$ is _____.
3. $\int_a^b \frac{dx}{\sqrt{1+x^2}}$ where $a = \frac{e-e^{-1}}{2}$ & $b = \frac{e^2-e^{-2}}{2}$
4. If $f(\pi) = 2$ & $\int_0^\pi (f(x) + f''(x)) \sin x dx = 5$, then find $f(0)$

Subjective Type Questions:

Evaluate the following definite integrals.

5. (i) $\int_0^1 \frac{dx}{x^2 + 2x \cos \alpha + 1}$ where $-\pi < \alpha < \pi$ (ii) $\int_0^1 \frac{x^2 - 1}{1 + x^2 + x^4} dx$
- (iii) $\int_0^{\pi/4} \frac{\sin \theta + \cos \theta}{9 + 16 \sin 2\theta} d\theta$
6. (i) $\int_{-1}^1 \frac{x dx}{\sqrt{5-4x}}$ (ii) $\int_0^{\pi/2} \frac{\cos x dx}{(1 + \sin x)(2 + \sin x)}$
- (iii) $\int_0^{\pi/4} \cos 2x \sqrt{1 - \sin 2x} dx$
7. (i) $\int_a^b \frac{|x|}{x} dx$ (ii) $\int_1^e \left(\frac{1}{\sqrt{x} \ln x} + \sqrt{\frac{\ln x}{x}} \right) dx$ (iii) $\int_2^e \left(\frac{1}{\ln x} - \frac{1}{\ln^2 x} \right) dx$
8. (i) $\int_0^{\pi/2} \frac{dx}{5 + 4 \sin x}$ (ii) $\int_2^3 \frac{dx}{(x-1)\sqrt{x^2-2x}}$ (iii) $\int_0^{\pi/2} \frac{dx}{1 + \cos \theta \cdot \cos x} \theta \in (0, \pi)$
9. (i) $\int_0^\pi \theta \sin^2 \theta \cos \theta d\theta$ (ii) $\int_0^{\pi/2} \frac{x + \sin x}{1 + \cos x} dx$ (iii) $\int_0^{3\pi/4} ((1+x)\sin x + (1-x)\cos x) dx$

10. (i) $\int_0^{\pi} \left[\cos^2 \left(\frac{3\pi}{8} - \frac{x}{4} \right) - \cos^2 \left(\frac{11\pi}{8} + \frac{x}{4} \right) \right] dx$
 (ii) $\int_0^{\pi/2} \sin \phi \cos \phi \sqrt{a^2 \sin^2 \phi + b^2 \cos^2 \phi} d\phi \quad a \neq b \quad (a > 0, b > 0)$
 (iii) $\int_2^3 \frac{dx}{\sqrt{(x-1)(5-x)}}$
11. $\int_{\pi/2}^{\pi} x^{\sin x} (1 + x \cos x \cdot \ln x + \sin x) dx$
12. If $f(x) = \begin{cases} 2 & 0 \leq x \leq 1 \\ x + [x] & 1 \leq x < 3 \end{cases}$, where $[.]$ denotes the greatest integer function. Evaluate $\int_0^2 f(x) dx$

Answer Key

- | | | | |
|--|---|------------------------------------|--|
| 1. (A) | 2. (5) | 3. (1) | 4. (3) |
| 5. (i) $\frac{\alpha}{2 \sin \alpha}$ if $\alpha \neq 0$; $\frac{1}{2}$ if $\alpha = 0$ | (ii) $-\frac{1}{2} \ln 3$ | (iii) $\frac{1}{20} \ln 3$ | |
| 6. (i) $\frac{1}{6}$ | (ii) $\ln \frac{4}{3}$ | (iii) $\frac{1}{3}$ | |
| 7. (i) $ b - a $ | (ii) $2\sqrt{e}$ | (iii) $e - \frac{2}{\ln 2}$ | |
| 8. (i) $\frac{2}{3} \tan^{-1} \frac{1}{3}$ | (ii) $\frac{\pi}{3}$ | (iii) $\frac{\theta}{\sin \theta}$ | |
| 9. (i) $-\frac{4}{9}$ | (ii) $\frac{\pi}{2}$ | (iii) $2(\sqrt{2} + 1)$ | |
| 10. (i) $\sqrt{2}$ | (ii) $\left(\frac{a^3 - b^3}{a^2 - b^2} \right) \frac{1}{3}$ | (iii) $\frac{\pi}{6}$ | 11. $\left(\pi - \frac{\pi^2}{4} \right)$ 12. $\frac{9}{2}$ |



Definite Integration: RACE-02

Single Correct Questions :

1. $\int_1^3 \frac{x^2}{x^2 - 4x + 8} dx$ is equal to -
 (A) 2 (B) 1 (C) $2 \tan^{-1} \frac{1}{2}$ (D) $2 \left(1 + \tan^{-1} \frac{1}{2} \right)$
2. The value of $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left(\frac{1 + \sin^2 x}{1 + \pi^{\sin x}} \right) dx$ is
 (A) $\frac{\pi}{2}$ (B) $\frac{5\pi}{4}$ (C) $\frac{3\pi}{4}$ (D) $\frac{3\pi}{2}$

Integer Type Questions:

3. Suppose that f, f' and f'' are continuous on $[0, \ln 2]$ and that $f(0) = 0, f'(\ln 2) = 6, f'(\ln 2) = 4$ and $\int_0^{\ln 2} e^{-2x} \cdot f(x) dx = 3$. Find the value of $\int_0^{\ln 2} e^{-2x} \cdot f''(x) dx$.
4. Suppose that the function f, g, f' and g' are continuous over $[0, 1]$, $g(x) \neq 0$ for $x \in [0, 1]$, $f(0) = 0, g(0) = \pi, f(1) = \frac{2009}{2}$ and $g(1) = 1$.

Find the value of the definite integral, $\int_0^1 \frac{f(x) \cdot g'(x) \{g^2(x) - 1\} + f'(x) \cdot g(x) \{g^2(x) + 1\}}{g^2(x)} dx$.

Subjective Type Questions:

Evaluate :

5. Evaluate : $\int_{-\sqrt{2}}^{\sqrt{2}} \frac{2x^7 + 3x^6 - 10x^5 - 7x^3 - 12x^2 + x + 1}{x^2 + 2} dx$
6. Evaluate : $\int_{-1}^1 \frac{(2x^{332} + x^{998} + 4x^{1668} \cdot \sin x^{691})}{1 + x^{666}} dx$
7. (i) $\int_0^3 |x^2 - x - 2| dx$ (ii) $\int_0^{\pi/2} |\sin x - \cos x| dx$ (iii) $\int_0^{\pi/2} \sqrt{\frac{1 - \sin 2x}{1 + \sin 2x}} dx$
8. $\int_0^1 x f''(x) dx$, where $f(x) = \cos(\tan^{-1} x)$
9. Given $f'(x) = \frac{\cos x}{x}$, $f\left(\frac{\pi}{2}\right) = a$, $f\left(\frac{3\pi}{2}\right) = b$. Find the value of the definite integral $\int_{\pi/2}^{3\pi/2} f(x) dx$.

10. (i) $\int_{-1}^1 \left(\frac{d}{dx} \left(\frac{1}{1+e^{1/x}} \right) \right) dx$
 (ii) $\int_{\ln 2}^{\ln 3} f(x) dx$, where $f(x) = e^{-x} + 2e^{-2x} + 3e^{-3x} + \dots \infty$
11. $\int_{\pi/6}^{\pi/3} \frac{dx}{1+\tan^5 x}$
12. $\int_0^3 \sqrt{\frac{x}{3-x}} dx$

Answer Key

- | | | | |
|---|-------------------------|--------------------|------------------------------|
| 1. (A) | 2. (C) | 3. (13) | 4. (2009) |
| 5. $\frac{\pi}{2\sqrt{2}} - \frac{16\sqrt{2}}{5}$ | 6. $\frac{\pi+4}{666}$ | | |
| 7. (i) $\frac{31}{6}$ | (ii) $2(\sqrt{2}-1)$ | (iii) $\ln 2$ | 8. $1 - \frac{3}{2\sqrt{2}}$ |
| 9. $2 - \frac{\pi}{2}(a-3b)$ | 10. (i) $\frac{2}{1+e}$ | (ii) $\frac{1}{2}$ | 11. $\frac{\pi}{12}$ |
| 12. $\frac{3\pi}{2}$ | | | |



Definite Integration: RACE-03

Single Correct Questions :

- $\int_6^{16} \frac{\log_e x^2}{\log_e x^2 + \log_e (x^2 - 44x + 484)} dx$ is equal to :
 (A) 6 (B) 8 (C) 5 (D) 10
- $\int_0^{\pi/2n} \frac{dx}{1 + \tan^n nx} =$
 (A) 0 (B) $\frac{\pi}{4n}$ (C) $\frac{n\pi}{4}$ (D) $\frac{\pi}{2n}$
- If $\int_0^{\pi} \frac{x^3 \cos^4 x \sin^2 x dx}{(\pi^2 - 3\pi x + 3x^2)} = k \int_0^{\pi/2} \sin^2 x dx$, then the value of k is :-
 (A) $\frac{\pi}{12}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{16}$ (D) $\frac{\pi}{8}$
- If $g(x) = \int_0^x \cos^4 t dt$, then $g(x + \pi)$ equals
 (A) $g(x) + g(\pi)$ (B) $g(x) - g(\pi)$ (C) $g(x) g(\pi)$ (D) $[g(x)/g(\pi)]$
- The value of $\int_{-1}^1 x^2 e^{[x^3]} dx$, where $[t]$ denotes the greatest integer $\leq t$, is :
 (A) $\frac{e-1}{3e}$ (B) $\frac{e+1}{3}$ (C) $\frac{e+1}{3e}$ (D) $\frac{1}{3e}$

Integer Type Questions

- The value of $I = \left| \int_{-3}^3 \ln \left(\frac{(\sqrt{1+x+x^2}(x-[x]))}{(\sqrt{1-x+x^2})} \right) dx \right|$ is (where $[.]$ denotes greatest integer function)

Subjective Type Questions :

- $\int_{1/3}^3 \frac{\sin^{-1} \frac{x}{\sqrt{1+x^2}}}{x} dx$
- (i) $\int_0^{\pi} \frac{(ax+b) \sec x \tan x}{4 + \tan^2 x} dx (a, b > 0)$
 (ii) $\int_0^{\pi} \frac{(2x+3) \sin x}{(1 + \cos^2 x)} dx$

9. Prove that :

(a)
$$\int_{\alpha}^{\beta} \sqrt{\frac{x-\alpha}{\beta-x}} dx = (\beta-\alpha) \frac{\pi}{2}$$

(b)
$$\int_{\alpha}^{\beta} \frac{x \cdot dx}{\sqrt{(x-\alpha)(\beta-x)}} = (\alpha+\beta) \frac{\pi}{2}, \text{ where } \alpha < \beta$$

10.
$$\int_0^{\pi/2} \ln(\sin^2 x \cos x) dx$$

11.
$$\int_{-\pi/2}^{\pi/2} \sqrt{\cos x - \cos^3 x} dx$$

12.
$$\int_{-1.5}^{10} \{2x\} dx, \text{ where } \{.\} \text{ denotes fractional part of } x.$$

Answer Key

Question	1	2	3	4	5	6
Answer	C	B	D	A	C	6

7.
$$\frac{\pi \ln 3}{2}$$

8. (i)
$$\frac{(a\pi+2b)\pi}{3\sqrt{3}}$$

(ii)
$$\frac{\pi(\pi+3)}{2}$$

10.
$$-\left(\frac{3\pi}{2}\right) \ln 2$$

11.
$$\frac{4}{3}$$

12.
$$\frac{23}{4}$$



Definite Integration: RACE-04

Single Correct Questions :

- If the value of the integral $\int_0^5 \frac{x + [x]}{e^{x-[x]}} dx = \alpha e^{-1} + \beta$, where $\alpha, \beta \in \mathbb{Q}$, $5\alpha + 6\beta = 0$, and $[x]$ denotes the greatest integer less than or equal to x ; then the value of $(\alpha + \beta)^2$ is equal to :
(A) 100 (B) 25 (C) 16 (D) 36
- If $f(x)$ is an even function which is also periodic with the period T and $\int_0^a f(x) dx = 3$ and $\int_{-T/2}^{3T/2} f(x) dx = 18$, then $\int_{-a}^{a+5T} f(x) dx$ is equal to -
(A) 96 (B) 93 (C) 51 (D) 48
- Let $f: (a, b) \rightarrow \mathbb{R}$ be twice differentiable function such that $f(x) = \int_a^x g(t) dt$ for a differentiable function $g(x)$. If $f(x) = 0$ has exactly five distinct roots in (a, b) , then $g(x)g'(x) = 0$ has at least :
(A) twelve roots in (a, b) (B) five roots in (a, b)
(C) seven roots in (a, b) (D) three roots in (a, b)
- Let $y = f(x)$ be a differentiable curve satisfying $\int_2^x f(t) dt = \frac{x^2}{2} + \int_x^2 t^2 f(t) dt$, then $\int_{-\pi/4}^{\pi/4} \frac{f(x) + x^9 - x^3 + x + 1}{\cos^2 x} dx$ equals -
(A) 0 (B) 1 (C) 2 (D) 4
- The value of $\lim_{n \rightarrow \infty} \sum_{r=0}^{2n-1} \frac{n}{n^2 + 4r^2}$ is :
(A) $\frac{1}{2} \tan^{-1}(2)$ (B) $\frac{1}{2} \tan^{-1}(4)$ (C) $\tan^{-1}(4)$ (D) $\frac{1}{4} \tan^{-1}(4)$
- If $L = \lim_{n \rightarrow \infty} \left(\frac{n+2}{n^2+n+1} + \frac{n+4}{n^2+2n+4} + \frac{n+6}{n^2+3n+9} + \dots + \frac{5}{7n} \right)$, then e^L is equal to -
(A) 1 (B) $\frac{1}{7}$ (C) 7 (D) e

Integer Type Question :

- If $\lambda = \lim_{n \rightarrow \infty} \frac{\left(\sum_{r=1}^n r \right) \left(\sum_{r=1}^n r^2 \right) \left(\sum_{r=1}^n r^3 \right)}{\sum_{r=1}^n r^8}$, then the value of 8λ is -

8. If $\int_0^{\pi/4} \sin^6 x \cos^4 x dx = \frac{3\pi}{1024} - \frac{K}{960}$, then K is equal to

Subjective Type Questions:

9. $\int_{20\pi+\frac{\pi}{6}}^{20\pi+\frac{\pi}{3}} (\sin x + \cos x) dx$

10. If $f(x) = \int_{1/x}^{\sqrt{x}} \sin t dt$, then find $f'(1)$.

11. Find the constants 'a' ($a > 0$) and 'b' such that $\lim_{x \rightarrow 0} \frac{\int_0^x \frac{t^2 dt}{\sqrt{a+t}}}{bx - \sin x} = 1$.

12. Prove the inequalities :

(i) $\frac{\pi}{6} < \int_0^1 \frac{dx}{\sqrt{4-x^2-x^3}} < \frac{\pi\sqrt{2}}{8}$

(ii) $2e^{-1/4} < \int_0^2 e^{x^2-x} dx < 2e^2$

(iii) $\frac{1}{3} < \int_0^1 x^{(\sin x + \cos x)^2} dx < \frac{1}{2}$

(iv) $\frac{1}{2} \leq \int_0^2 \frac{dx}{2+x^2} \leq \frac{5}{6}$

Answer Key

Question	1	2	3	4	5	6	7	8
Answer	B	C	C	C	B	C	3	3

9. $(\sqrt{3}-1)$

10. $\frac{3}{2} \sin 1$

11. $a = 4$ and $b = 1$